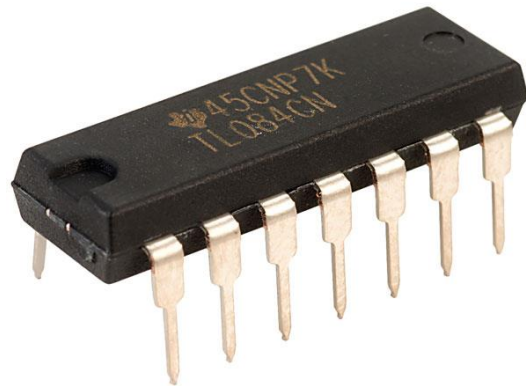


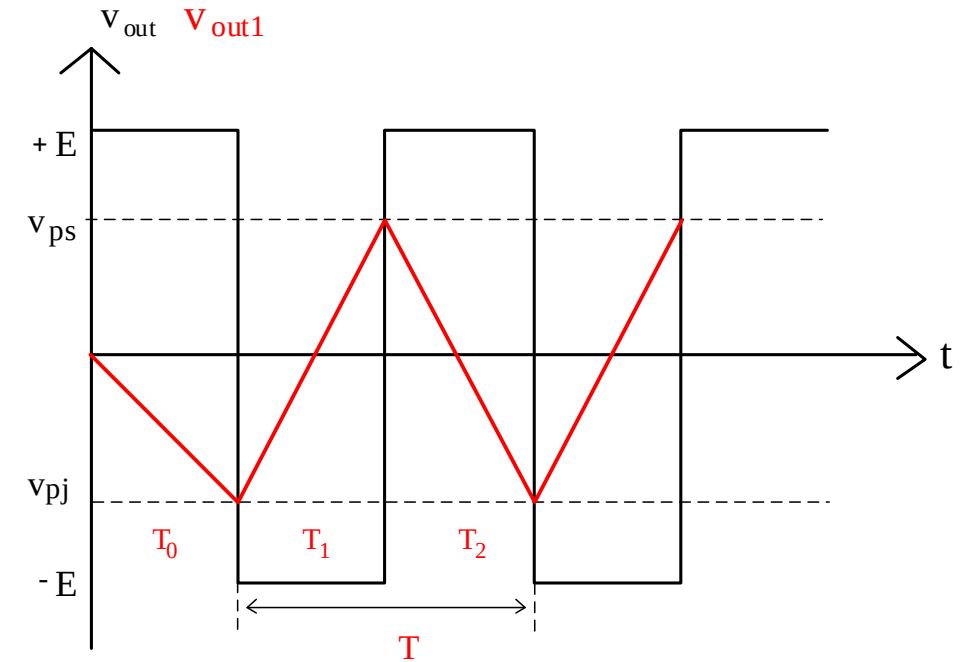
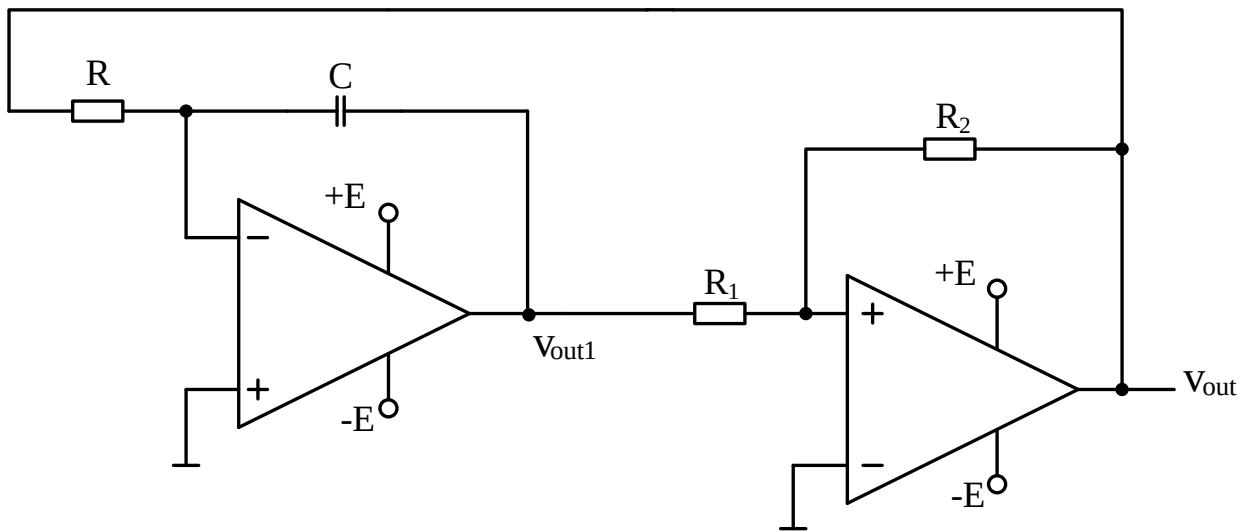
# Bazele Circuitelor Electronice

Aplicații cu Amplificatoare Operaționale

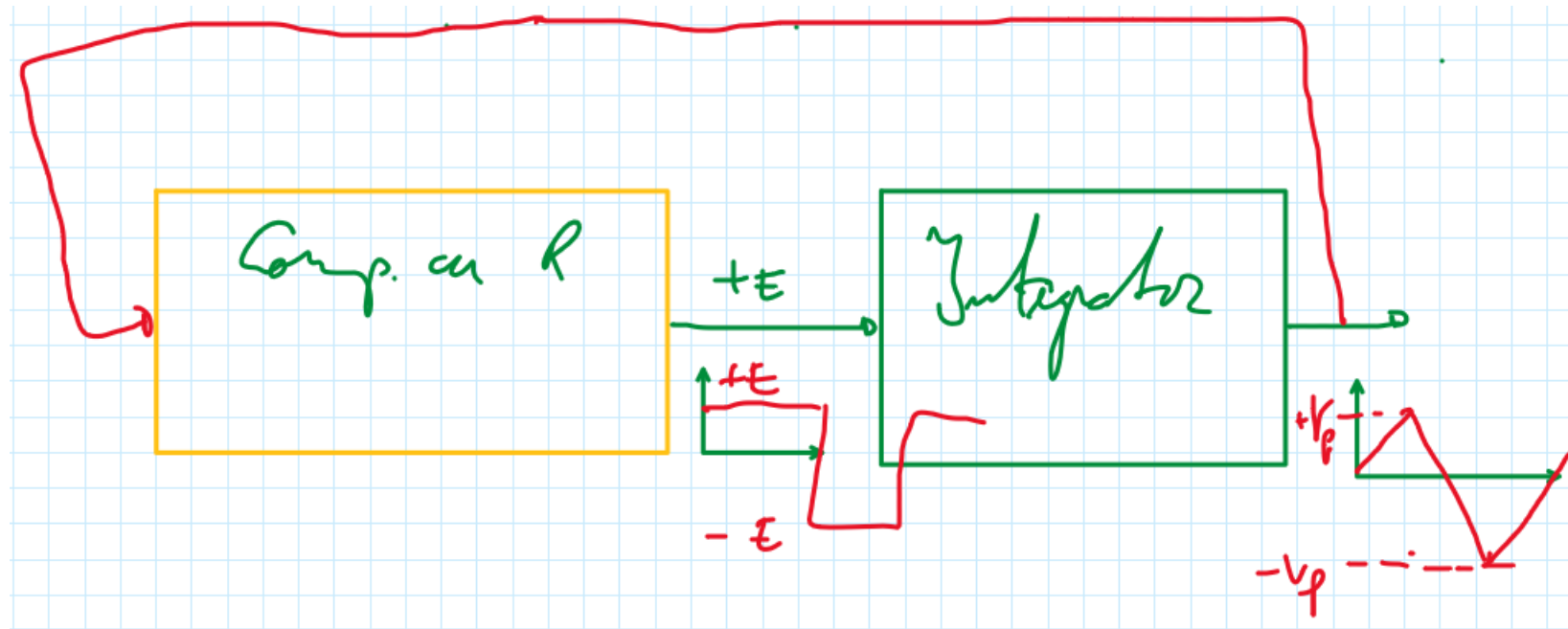


# Generatoare de semnal cu AO

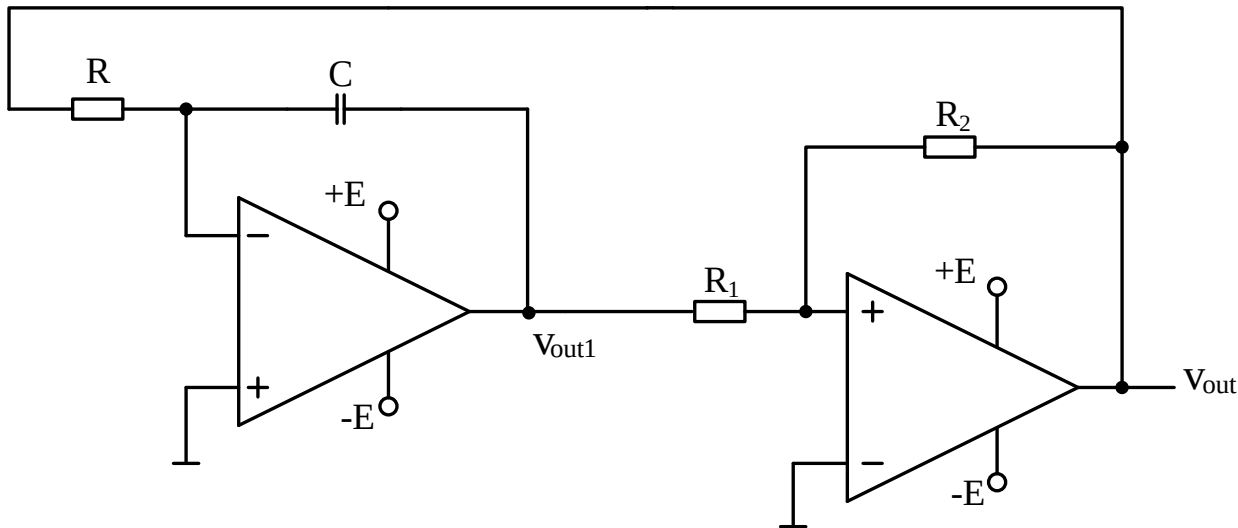
# Generator de semnale triunghiular și dreptunghiular



# Generator de semnale triunghiular și dreptunghiular



# Generator de semnale triunghiular și dreptunghiular

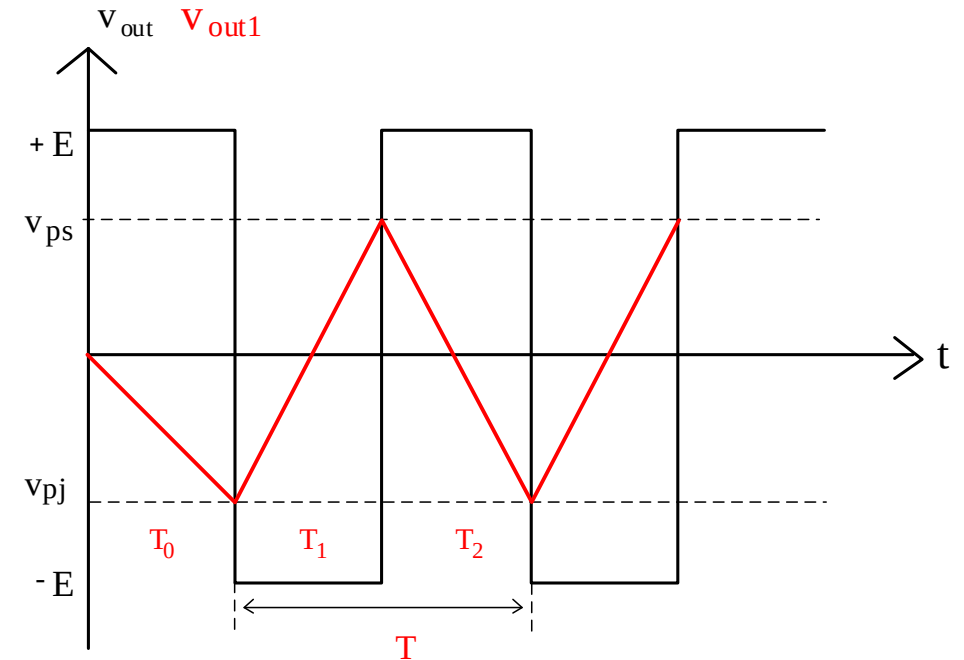


$$i_c = C \frac{dV_c}{dt} \Rightarrow v_c(t) = \frac{1}{C} \int i dt + V_c(0)$$

$$V^+ = V^-$$

$$V_c(t) = \frac{I}{C} t + V_c(0)$$

$$I = \frac{E}{R}$$



$$V^+ = V^- = \frac{\frac{V_P}{R_1} + \frac{V_0}{R_2}}{\frac{1}{R_1} + \frac{1}{R_2}} = 0 \Rightarrow V_P = -\frac{R_1}{R_2} V_{out}$$

# Generator de semnale triunghiular și dreptunghiular

$$V_C(t) = \frac{I}{C}t + V_C(0) \Rightarrow V_{out1}(t) = -\frac{I}{C}t + V_{out1}(0)$$

**$T_1$  – condiții inițiale:**

$$V_{out1}(0) = V_{pj} = -\frac{R_1}{R_2}E$$

$$V_{out1}(T_1) = V_{ps} = \frac{R_1}{R_2}E$$

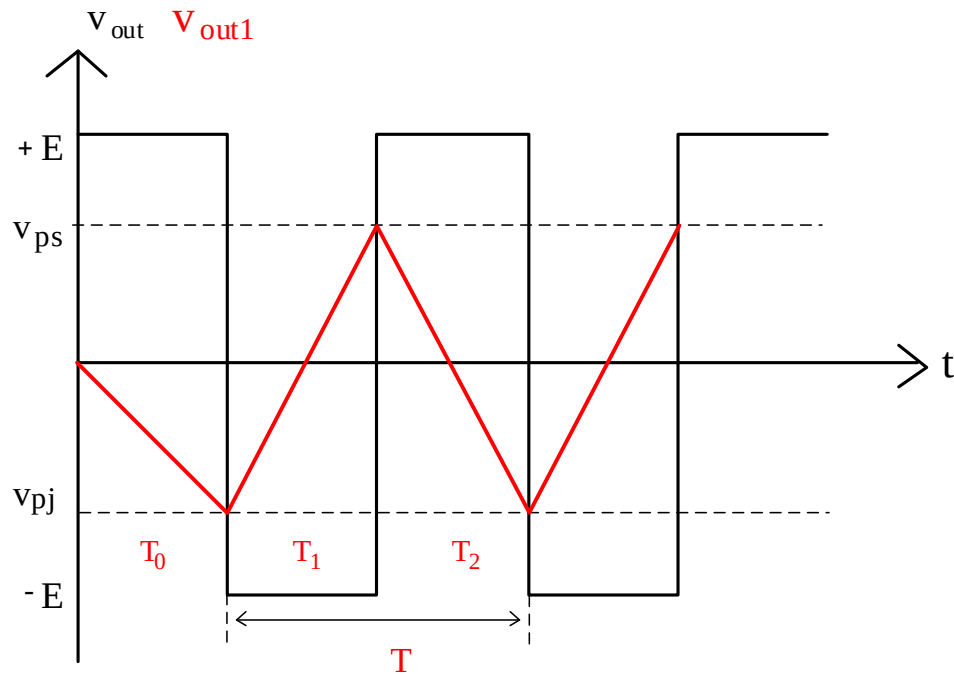
$$T_1 = 2 \cdot \frac{R_1}{R_2} \cdot R \cdot C$$

**$T_2$  - condiții inițiale:**

$$V_{out1}(0) = V_{ps} = \frac{R_1}{R_2}E$$

$$V_{out1}(T_2) = V_{pj} = -\frac{R_1}{R_2}E$$

$$T_2 = 2 \cdot \frac{R_1}{R_2} \cdot R \cdot C$$

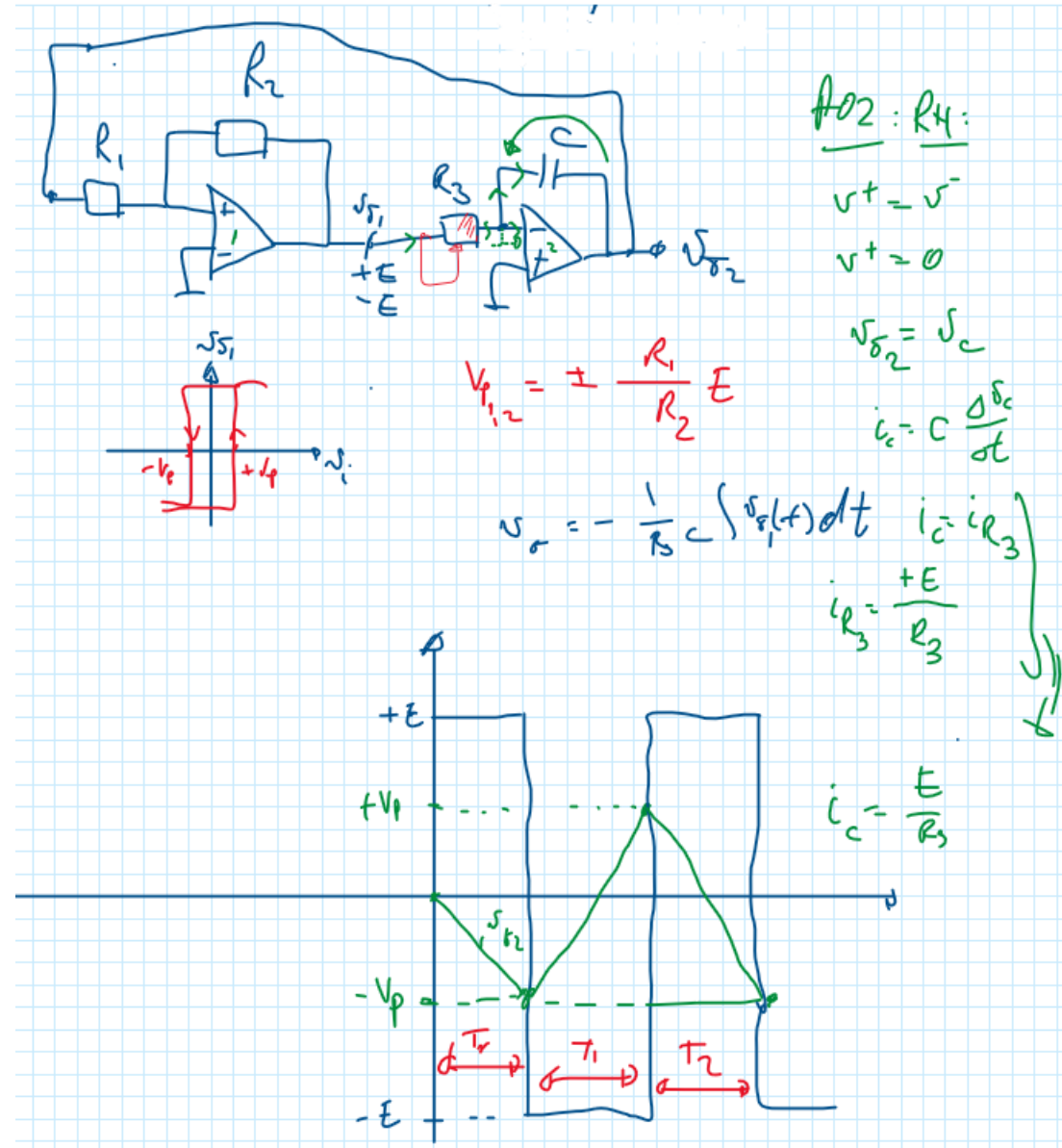


$$T = T_1 + T_2 \Rightarrow T = 4 \frac{R_1}{R_2} RC$$

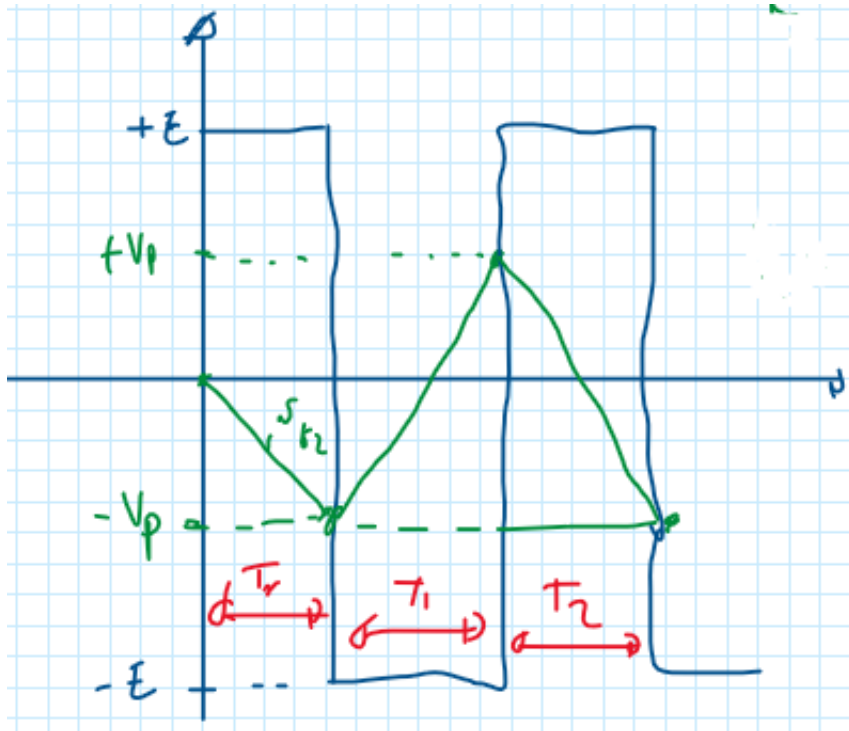
$$f_u = \frac{T_2}{T} = 50\%$$



# Generator de semnale triunghiular și dreptunghiular



# Generator de semnale triunghiular și dreptunghiular



$$T_0: \left. \begin{aligned} v_{\sigma_2} &= v_c \\ i_c &= -C \frac{\Delta v_c}{\Delta t} \\ i_c &= \frac{E}{R_3} \end{aligned} \right\}$$

$$\Rightarrow \frac{E}{R_3} = -C \frac{\Delta v_{\sigma_2}}{T_0}$$

$$\frac{E}{R_3} = -C \frac{-V_p - 0}{T_0}$$

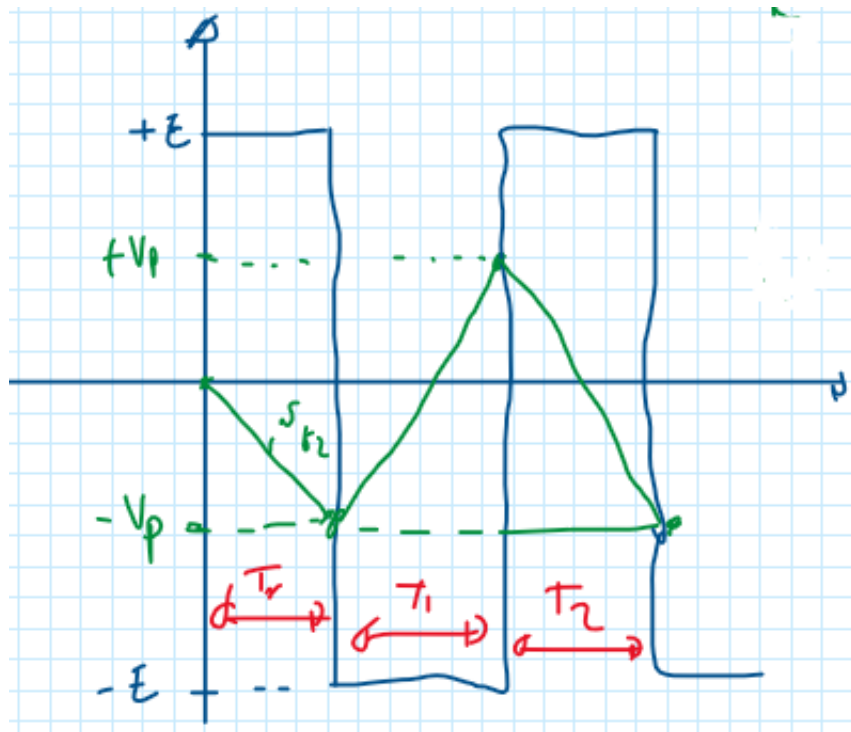
$$\frac{E}{R_3} = C \frac{V_p}{T_0}$$

$$\Rightarrow T_0 = R_3 \cdot C \frac{V_p}{E} \Rightarrow V_p = \frac{R_1}{R_2} E$$

$$T_0 = R_3 \cdot C \frac{R_1}{R_2}$$



# Generator de semnale triunghiular și dreptunghiular $T_1$ :



$$i_c = -C \frac{\Delta s_{k2}}{\Delta t}$$

$$i_c = -\frac{E}{R_3}$$

$\Rightarrow$

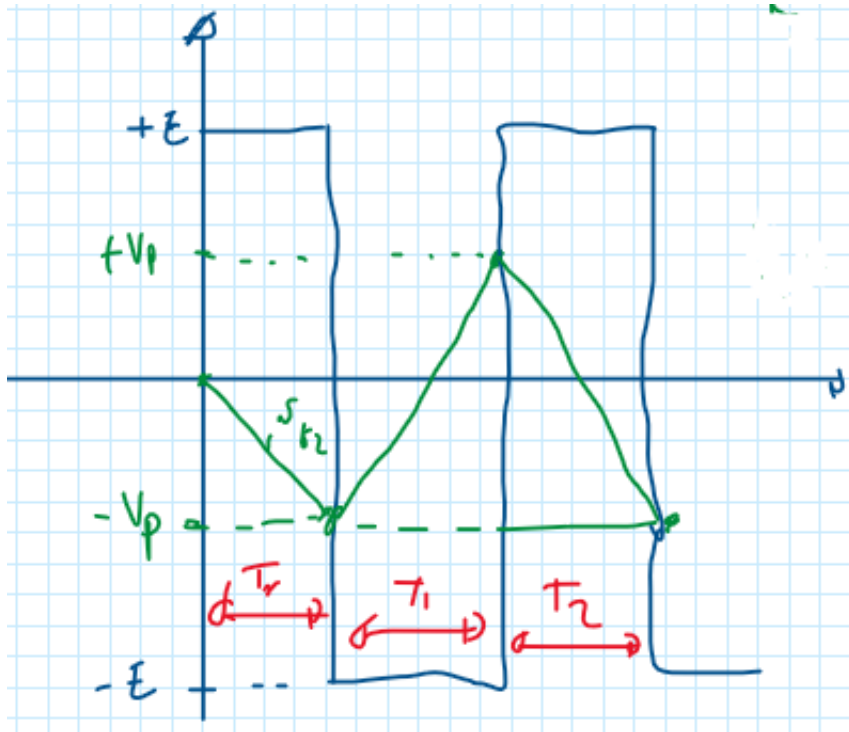
$$-\frac{E}{R_3} = -C \frac{V_p - (-V_p)}{T_1}$$

$$\frac{E}{R_3} = C \frac{2V_p}{T_1}$$

$$T_1 = 2R_3 C \frac{V_p}{E} \quad \Rightarrow V_p = \frac{R_1}{R_2} E$$

$$T_1 = 2R_3 C \frac{R_1}{R_2}$$

# Generator de semnale triunghiular și dreptunghiular $T_1$ :



$$T_1 = 2R_3 < \frac{R_1}{R_2}$$

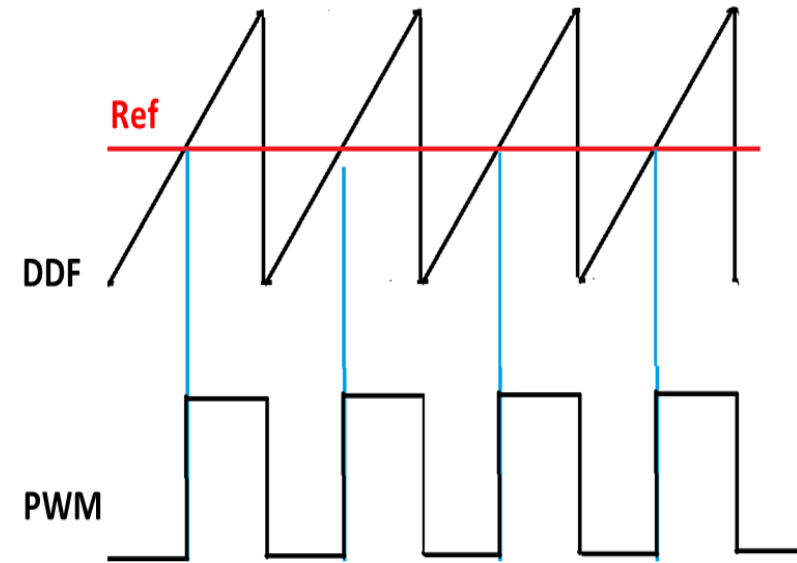
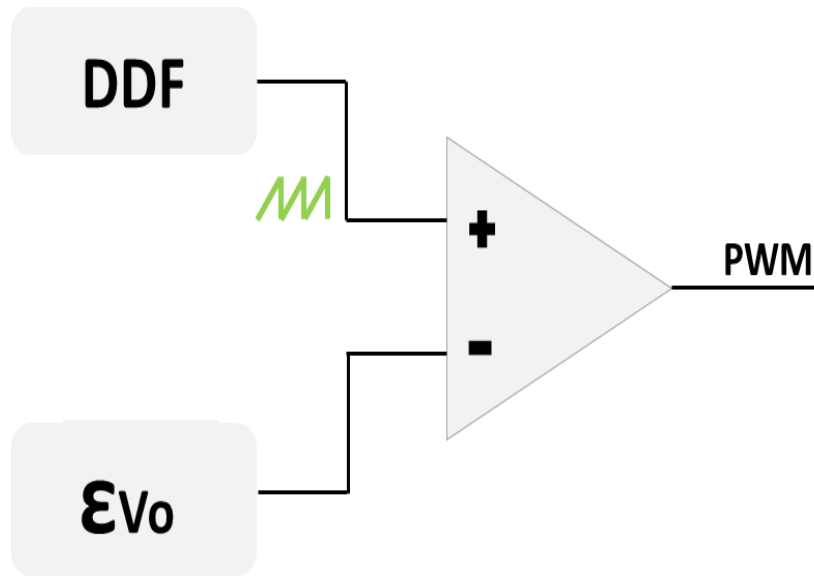
$$T_2 = T_1$$

$\Rightarrow$

$$T = 2T_1 = 4R_3 < \frac{R_1}{R_2}$$

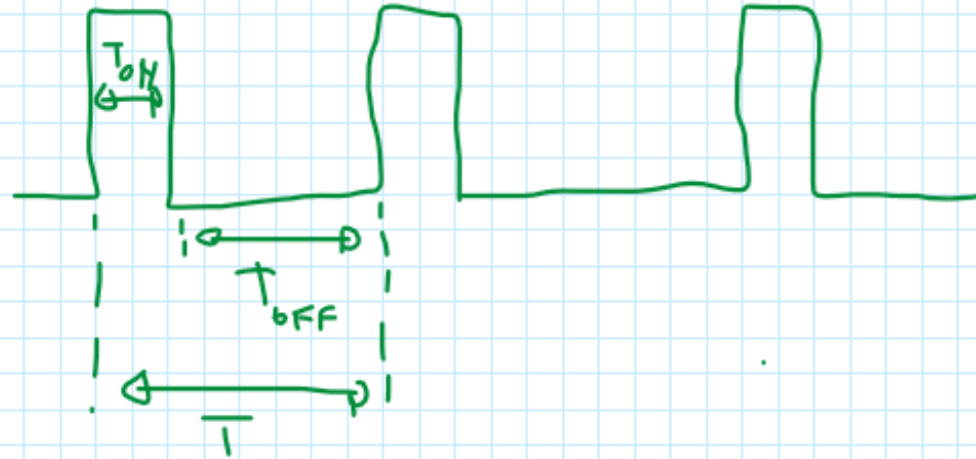
$$f = \frac{1}{T} = \frac{1}{4R_3} < \frac{R_2}{R_1}$$

# Generator PWM



# Generator PWM

PWM - Pulse Width Modulation

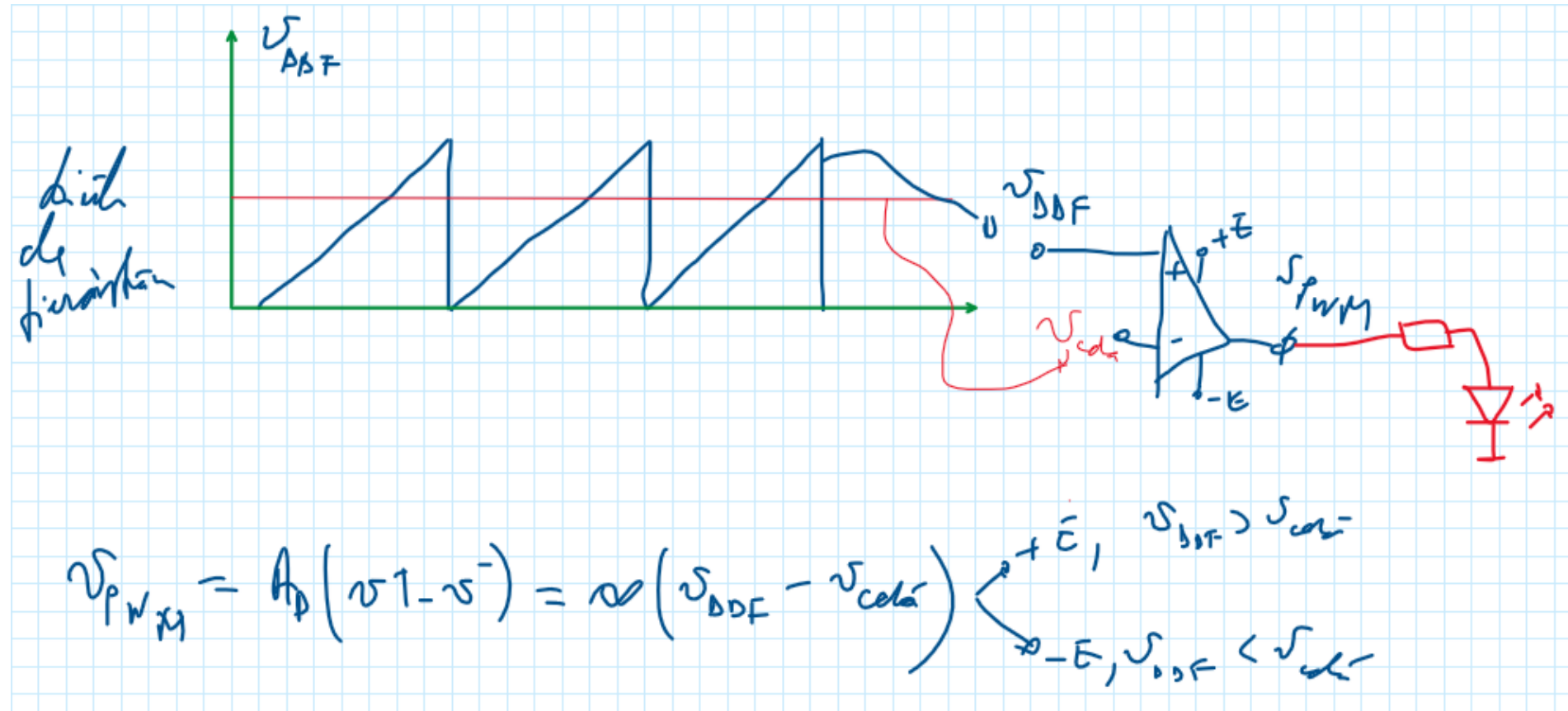


$$T = T_{ON} + T_{OFF}$$

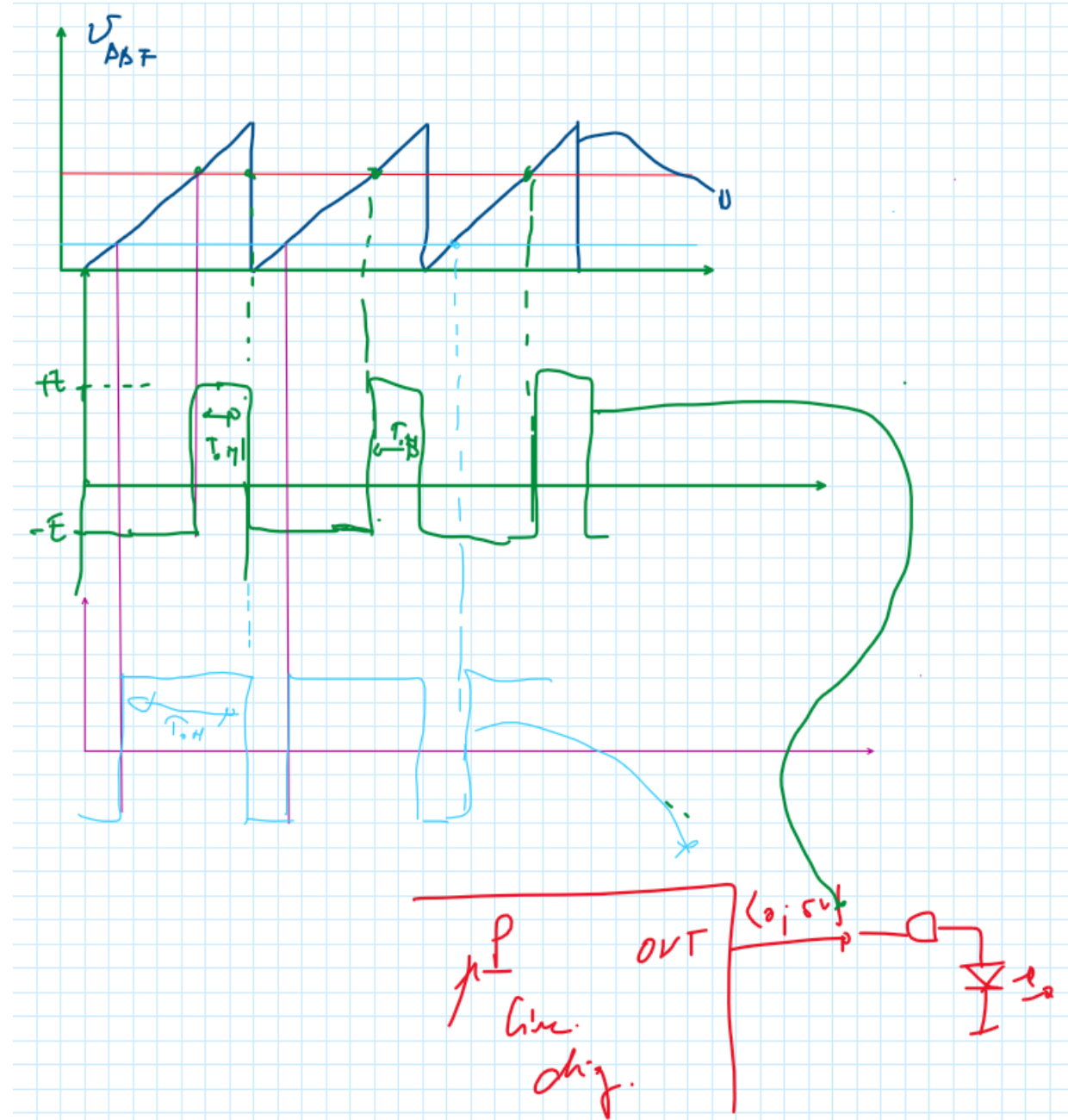
f. factor di segnale  $f_m = \frac{T_{ON}}{T} \cdot 100 [\%]$



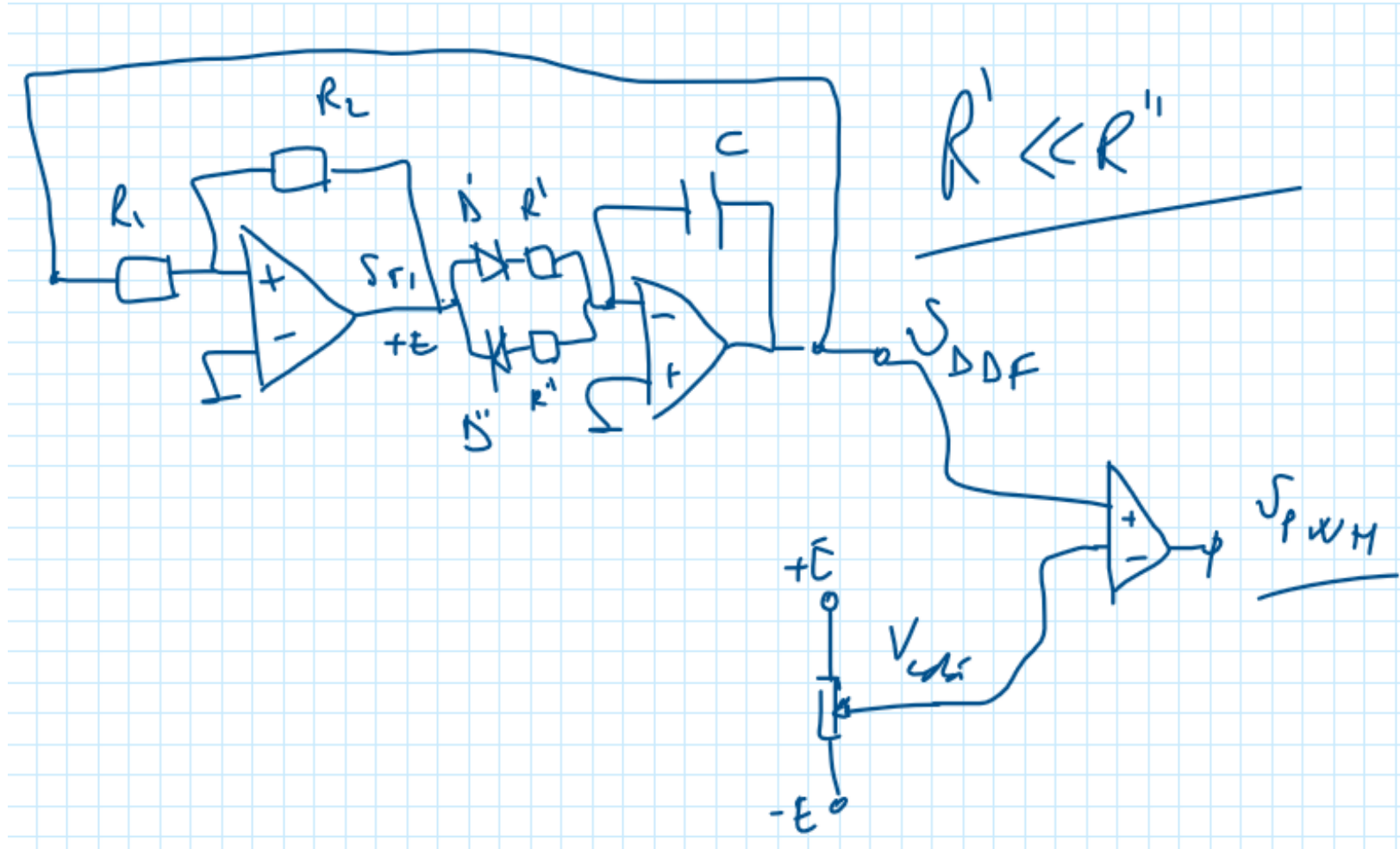
# Generator PWM



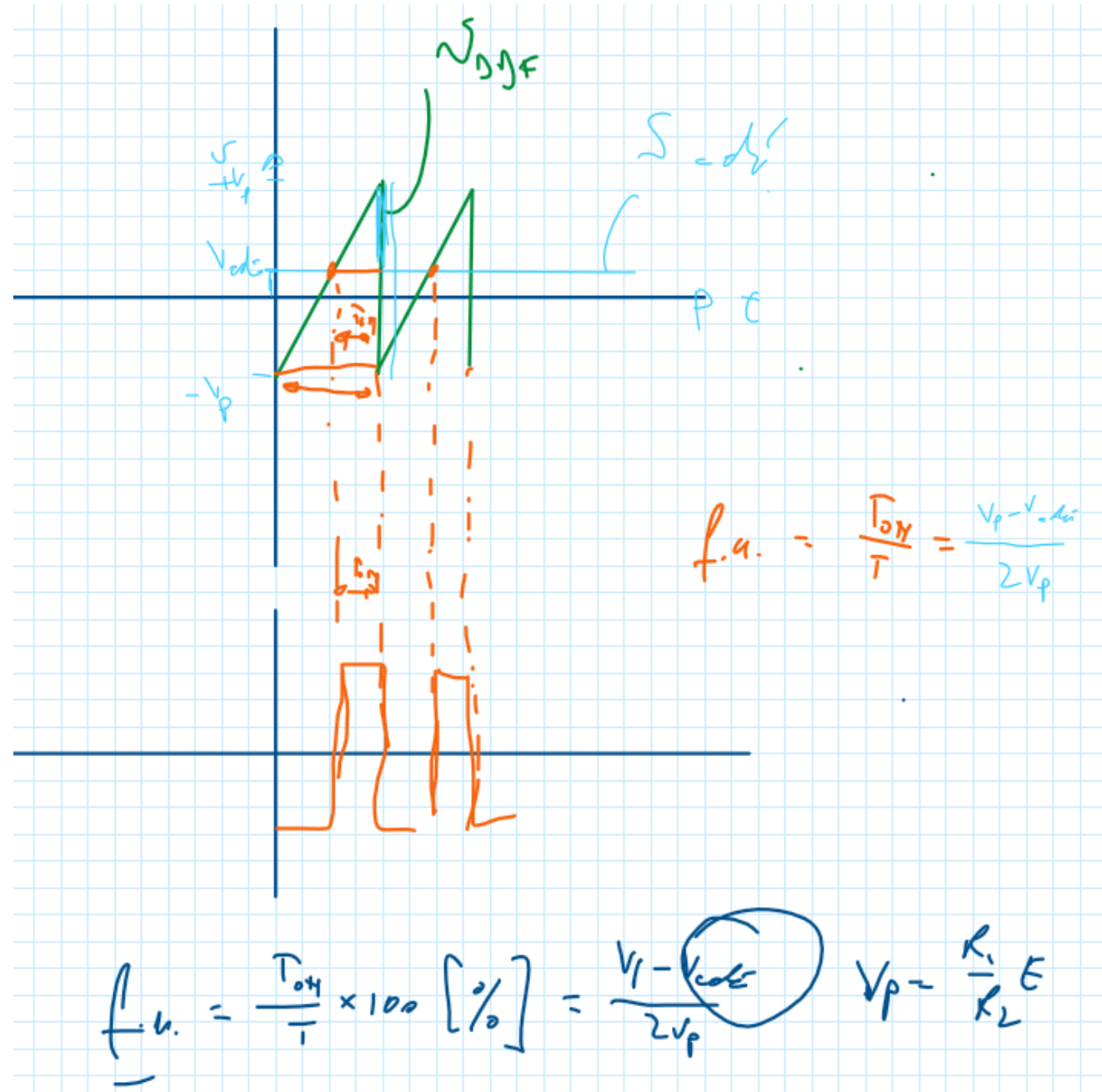
# Generator PWM



# Generator PWM

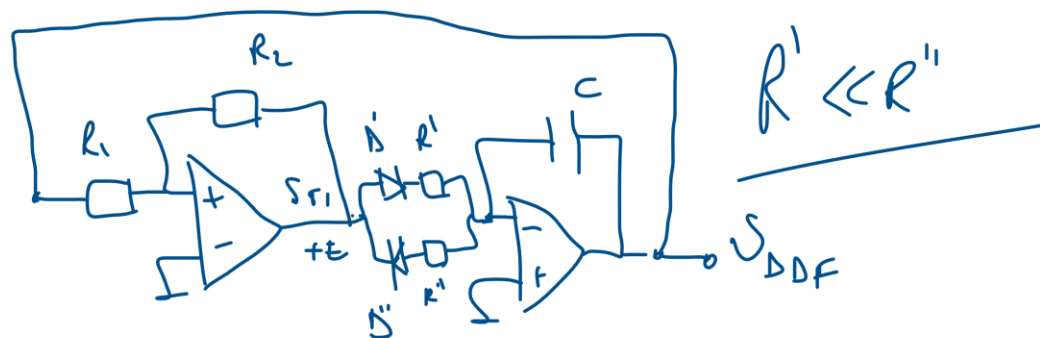
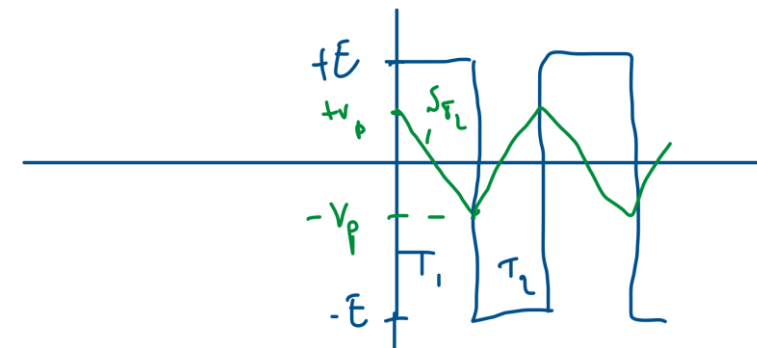


# Generator PWM



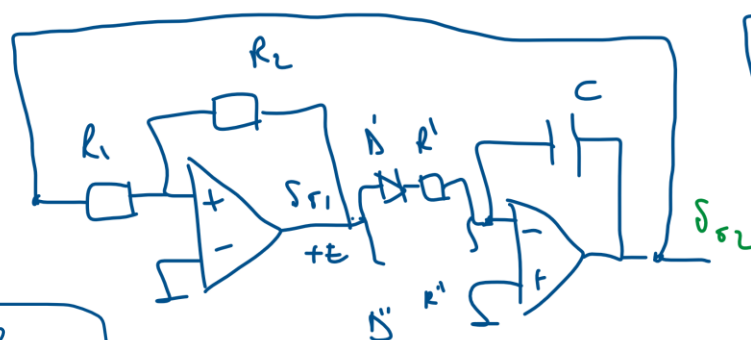


# Generator PWM

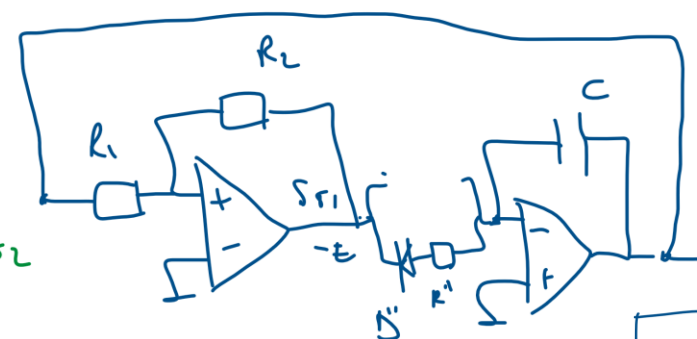


$$V_{FL} = +E \Rightarrow D' - ON \\ D'' - OFF$$

$$V_{FL} = -E, D' - OFF \\ D'' - ON$$



$$T_1 = 2R' \cdot C \cdot \frac{R_1}{R_2}$$



$$T_2 = 2R' \cdot C \cdot \frac{R_1}{R_2}$$

# Generator PWM

$$T_1 = 2R' \leq \frac{R_1}{R_2}$$

$$T_2 = 2R'' \leq \frac{R_1}{R_2}$$

$$T = T_1 + T_2 = 2(R' + R'') \leq \frac{R_1}{R_2}$$

$$f_{u.} = \frac{T_1}{T_1 + T_2} \times 100 = \frac{2R' \leq \frac{R_1}{R_2}}{2(R' + R'') \leq \frac{R_1}{R_2}} \times 100 = \frac{R'}{R' + R''} \times 100 [\%]$$

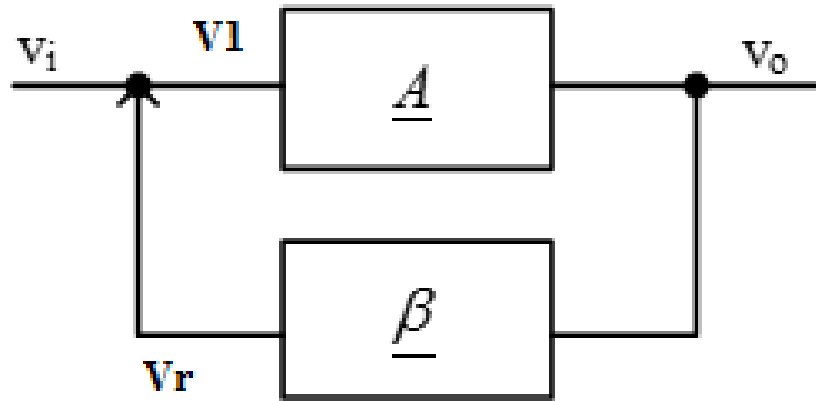
$$\text{pp. } R' \gg R'' \Rightarrow T_1 \gg T_2$$

$$\text{pp. } R' \ll R'' \Rightarrow T_2 \gg T_1$$

DDF



# Generator sinusoidal cu punte Wien



$$\underline{H} = \frac{V_o}{V_i} = \frac{\underline{A}}{1 - \underline{A}\underline{\beta}}$$



# Generator sinusoidal cu punte Wien

Condiția de oscilație (Barkhausen):

$$\underline{A} \cdot \underline{\beta} = 1$$



# Generator sinusoidal cu punte Wien

Condiția de oscilație (Barkhausen):

$$\underline{A} \cdot \underline{\beta} = 1$$

$$\underline{A} = |\underline{A}| e^{j\varphi_A}$$

$$\underline{\beta} = |\underline{\beta}| e^{j\varphi_\beta}$$



$$|\underline{A}| |\underline{\beta}| e^{j(\varphi_A + \varphi_\beta)} = 1$$

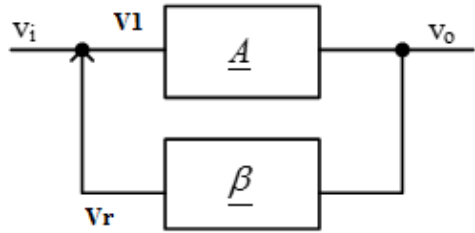
$$\left\{ \begin{array}{l} |\underline{A}| |\underline{\beta}| = 1 \\ e^{j(\varphi_A + \varphi_\beta)} = 1 \end{array} \right.$$

$$e^{j(\varphi_A + \varphi_\beta)} = 1 \longrightarrow \varphi_A + \varphi_\beta = 0 \quad (2k\pi)$$



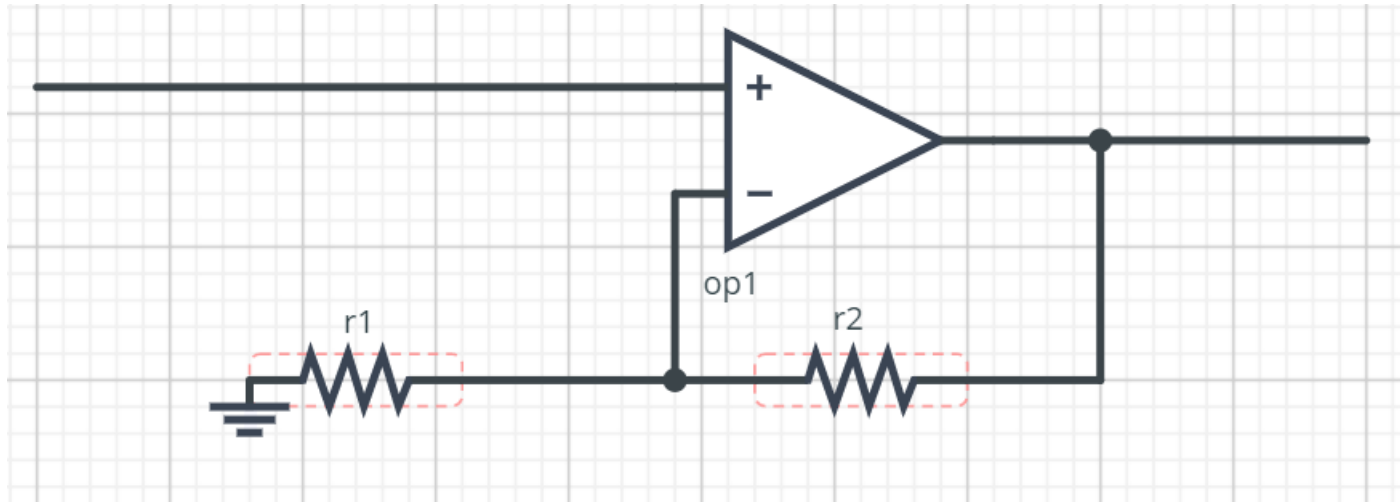
# Generator sinusoidal cu punte Wien

A:



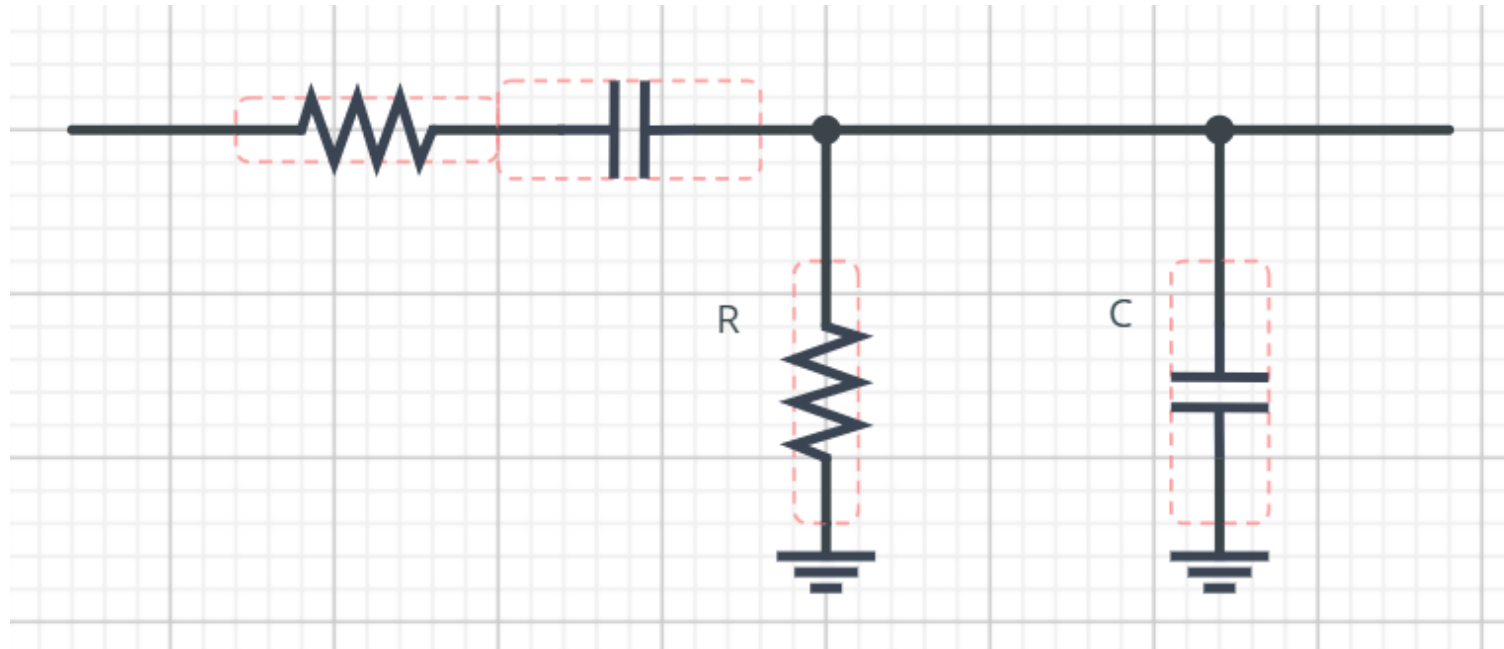
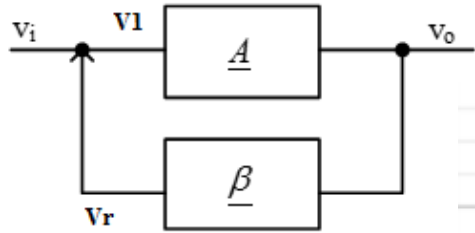
$$|A| = 1 + \frac{R_2}{R_1} = 3$$

$$\varphi_A = 0$$



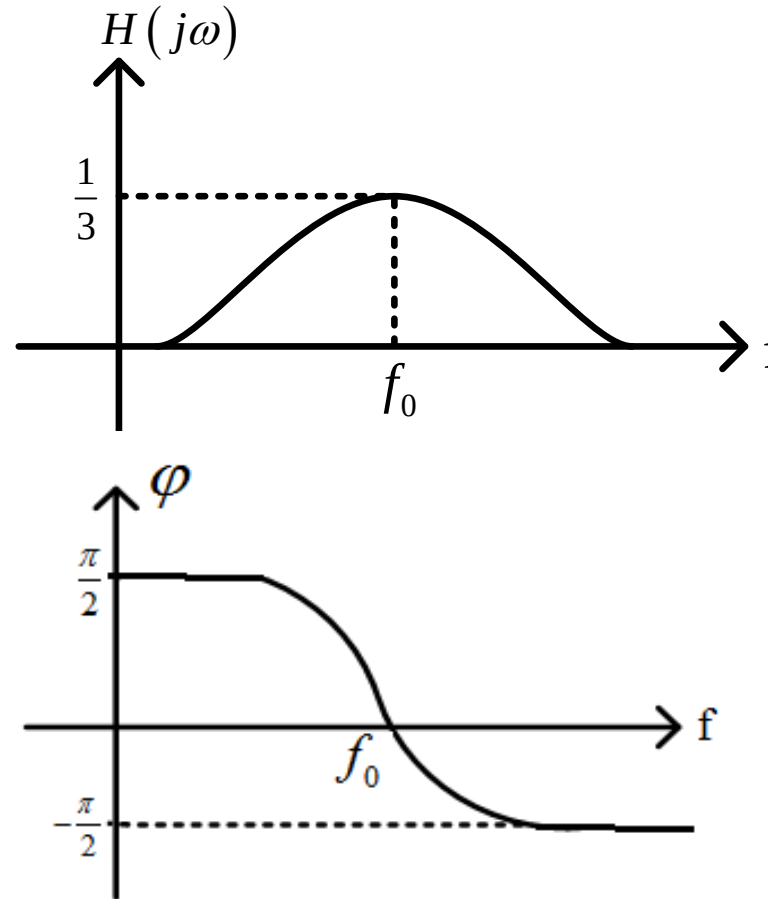
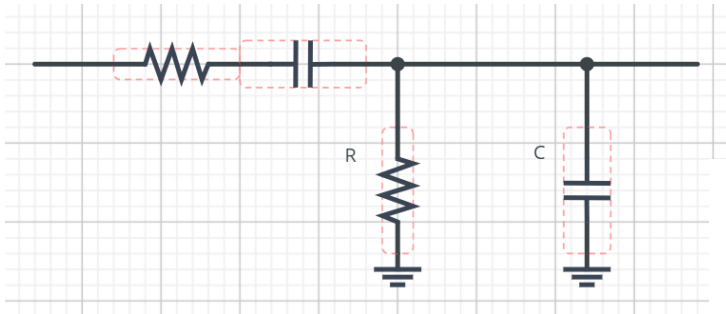
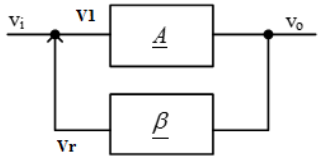
# Generator sinusoidal cu punte Wien

B:



# Generator sinusoidal cu punte Wien

**B:**

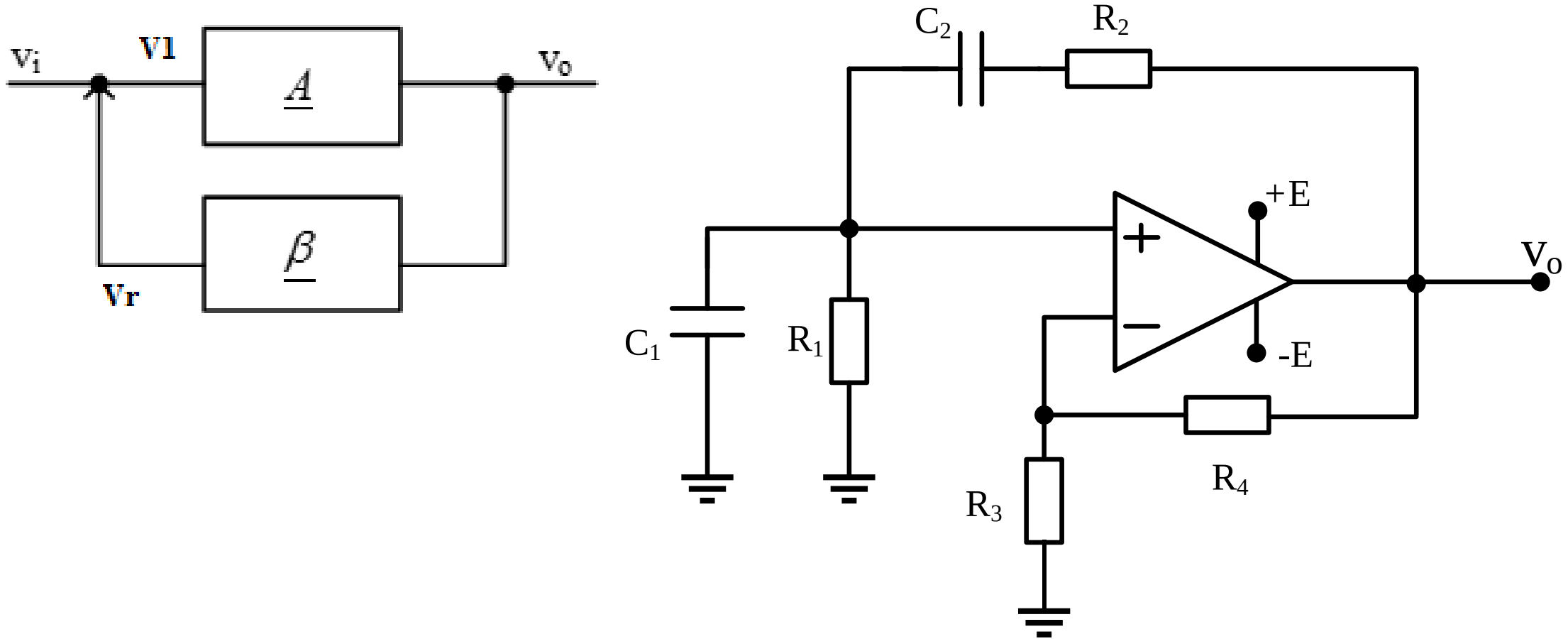


$$\begin{cases} \left| \frac{v_o}{v_i} \right| = \frac{1}{\sqrt{(1-x^2)^2 + 9x^2}} \\ \varphi = \frac{1-x^2}{3x} \end{cases}$$

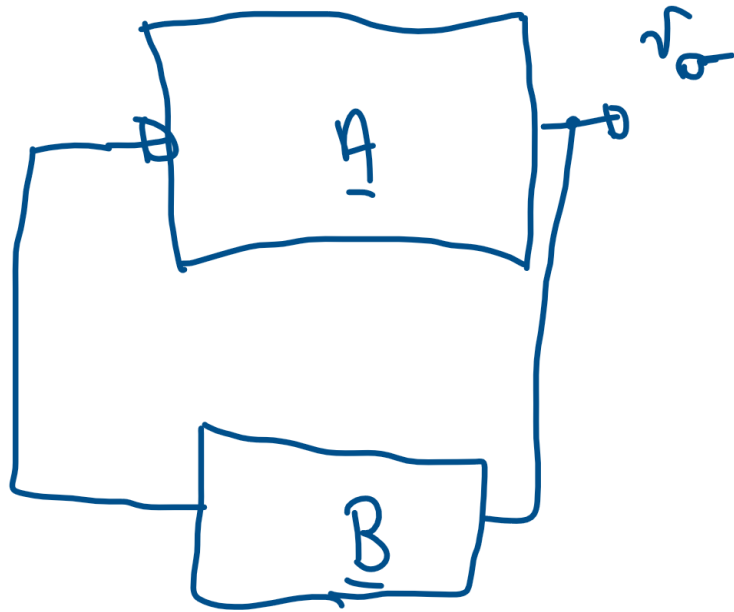




# Generator sinusoidal cu punte Wien



# Generator sinusoidal cu punte Wien



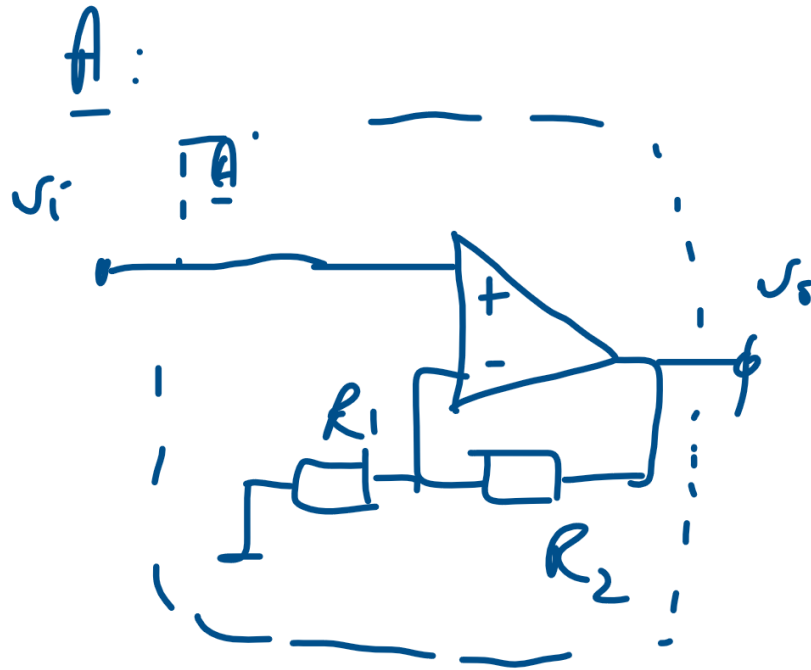
BARKHAUSEN

$$\underline{A} \times \underline{B} = 1$$

$$\Rightarrow \begin{cases} |A| \cdot |B| = 1 \\ \varphi_A + \varphi_B = 2k\pi \rightarrow 0 \end{cases}$$



# Generator sinusoidal cu punte Wien

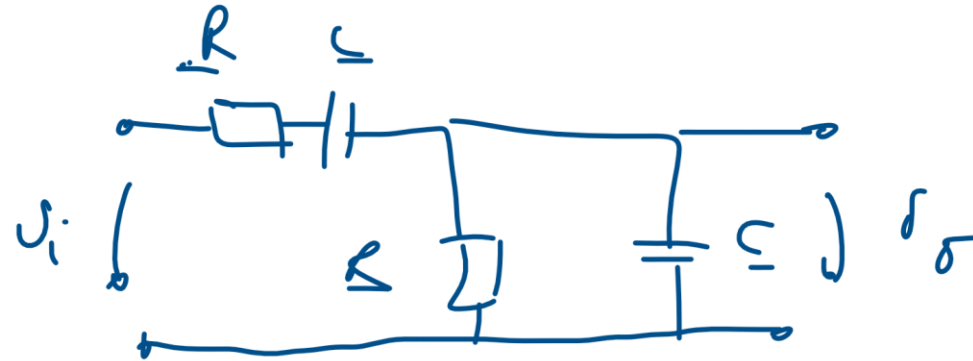


$$\left\{ \begin{array}{l} |H| = \left| \frac{s_o}{s_i} \right| = 1 + \frac{R_2}{R_1} \\ \varphi = 0 \end{array} \right.$$

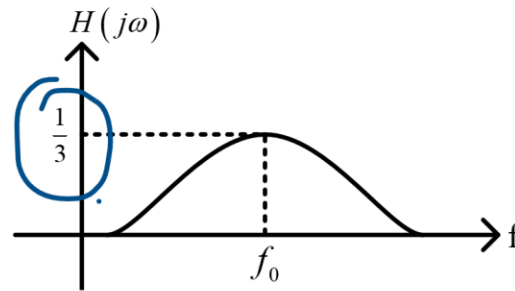


# Generator sinusoidal cu punte Wien

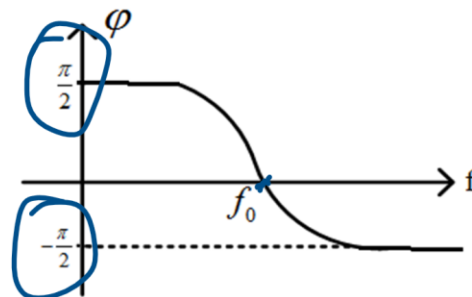
Wien (FTB)



$$\underline{B} = \underline{H} = \frac{\underline{U}_o}{\underline{U}_i}$$



$$f_0 = \frac{1}{2\pi RC}$$



$$|B| = \frac{1}{3}$$



# Generator sinusoidal cu punte Wien

$$|A| \cdot |B| = 1$$

$$\left(1 + \frac{R_2}{R_1}\right) \cdot \frac{1}{3} = 1$$

$$\frac{R_2}{R_1} = 2 \quad \Rightarrow \quad \underline{R_2 = 2R_1}$$



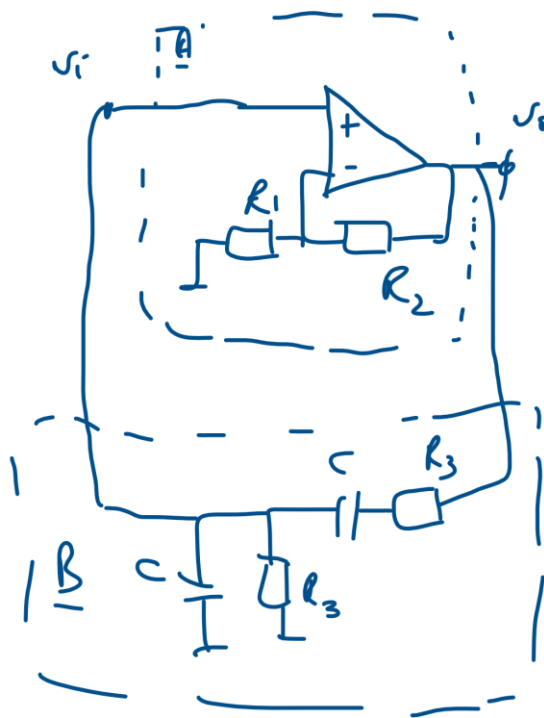
# Generator sinusoidal cu punte Wien

Cond. de  
oscilație

$$R_2 = 2R_1$$

frec. n. de  
oscilație

$$f = \frac{1}{2\pi R_3 C}$$



$$\begin{aligned} \varphi_A + \varphi_B &= 0 \\ -\frac{\pi}{2} < \varphi_B < \frac{\pi}{2} \\ \varphi_A &= 0 \\ \varphi_B &= 0 \end{aligned}$$

$$f = \frac{1}{2\pi R_3 C}$$

$$\begin{aligned} |A| \cdot |B| &= 1 \\ |A| &= 1 + \frac{R_2}{R_1} \\ |B| &= \frac{1}{3} \end{aligned} \Rightarrow \boxed{R_2 = 2R_1}$$